

DBMS FINAL PROJECT: DATABASE DESIGN AND IMPLEMENTATION FOR A WEB-BASED SYSTEM

I. Project Overview

As a continuation and collaboration of your Web Design project, this final requirement

For the DBMS course focuses on the database design, creation, and manipulation

Aspects of your web-based system.

You and your groupmates will design and implement the database component that

Supports your simple web system's functionality, applying your knowledge in database

Normalization, SQL queries, and relational design.

This project evaluates how well your database is structured, how efficiently it stores and

Retrieves data, and how it connects to your system through MySQL.

II. Project Requirements

1. Database Design

- a) Construct an Entity-Relationship Diagram using Chen Notation to represent all

Entities, attributes, and relationships.

- b) Apply Database Normalization up to Third Normal Form (3NF) to eliminate

Redundancy and ensure data integrity.

- c) Clearly define all primary keys, foreign keys, and data constraints such as NOT NULL,

UNIQUE, etc.

2. Documentation Contents

- a) Introduction – a brief overview of the system (what it is and what it does).
- b) Database Design Process – briefly explain how you designed your database before

Coding.

- Entity Identification – Describe each entity (e.g., User, Product, Appointment,

Etc.) and why it was necessary.

- Attribute Definition – List important attributes per entity with short

Descriptions.

- Relationships – Explain how the entities are connected (e.g., One-to-Many

Between Users and Appointments).

c) Entity–Relationship Diagram (ERD)

- Present your ERD using Chen notation.
- Include labels for relationships (e.g., Registers, Contains, Belongs to).
- Insert a clear image of the ERD here.

III. Database Implementation (MySQL)

1. Use MySQL to create the actual database for your system.
2. Implement all necessary tables, fields, and relationships based on your ERD.

IV. SQL Operations and Queries

Demonstrate mastery of SQL through queries used in your system and additional

Required examples.

1. The use of DML (Insert, Update, Delete, Select)
2. The use of any Data Query Language (DQL) that is applicable with your system.
 - a) WHERE, Logical Operators, Comparison Operators
 - b) Aggregate Functions, Group By, Having, Order By
 - c) Joins (Inner, Left, Right)
 - d) Use of Aliases, Limit, and Like Operator

V. Integration with the Web-Based System

1. Ensure that the PHP front-end system connects properly to the MySQL database.
2. All CRUD operations in the web system must correspond to actual SQL commands.

3. Queries executed through the system must reflect properly.

VI. Deliveries

1. Google Drive Folder Submission – All groups must submit their final project through a

Google Drive folder named: DBMS-FINALPROJECT-SystemName (example: DBMSFinalProject-StudentInformationManagementSystem)

A) Contents of the folder

- Working project files (HTML, CSS, JS, PHP)
- Database files
- Documentation

ANSWER:

I. Project Overview

The **E-Library Web-Based System** is a platform that allows users to manage, view, and publish books online. It provides functionalities for registering authors, adding books, categorizing them by genres, featuring books, and uploading associated files (PDF and cover images). Users can also browse books by genre, view featured books, and read them in a PDF viewer.

This DBMS project focuses on the **database design and implementation** supporting this system, demonstrating proper relational design, normalization, SQL operations, and integration with PHP for CRUD operations.

II. Project Requirements

1. Database Design

a) Entity-Relationship Diagram (ERD)

The system has the following main entities:

1. **Authors**
2. **Books**
3. **Genres** (*optional, can be a lookup table*)

Relationships:

- An **Author** can write many **Books** → One-to-Many.
- A **Book** belongs to one **Genre** → Many-to-One (if using a separate genres table).

Entities and Attributes:

Entity	Attributes
Authors	id (PK), name, email

Entity	Attributes
Books	id (PK), title, author_id (FK), genre, description, cover_image, bookfile, date_published, featured
Genres (optional)	id (PK), name

Constraints:

- Authors.email → UNIQUE, NOT NULL
- Books.title → NOT NULL
- Books.author_id → Foreign Key referencing Authors.id
- Books.genre → can be ENUM or FK if using Genres table

(Insert ERD image here – using Chen notation: rectangles for entities, ovals for attributes, diamonds for relationships)

b) Database Normalization

- **1NF:** Each attribute contains atomic values, no repeating groups.
- **2NF:** All non-key attributes are fully functional dependent on primary keys.
- **3NF:** No transitive dependencies; e.g., author info stored only in Authors table, not in Books.

This ensures minimal redundancy and data integrity.

c) Attribute Definition

Authors Table:

Attribute Description

id	Primary key, auto-incremented
name	Full name of the author
email	Email address, unique

Books Table:

Attribute	Description
id	Primary key, auto-incremented
title	Book title
author_id	Foreign key referencing Authors.id
genre	Book genre (Business, Technology, etc.)
description	Short description of the book
cover_image	Path to book cover image
bookfile	Path to PDF file
date_published	Date the book was published
featured	Boolean flag for featured books

d) Relationships

1. Authors → Books

- Type: One-to-Many
- Meaning: One author can write multiple books.

2. Books → Genres *(if using separate table)*

- Type: Many-to-One
- Meaning: Each book belongs to one genre.